

#1 Suppose the matrix A is $(n \times m)$ SVD of A : $A = U \Sigma V^T$.

Describe the three matrices on the right by listing their dimensions + other properties

$A: U \Sigma V^T$ let $A \in \mathbb{R}^{n \times m}$

let $A \in \mathbb{R}^{n \times m}$

— $U \in \mathbb{R}^{n \times n}$
orthogonal matrix $\Rightarrow U^T U = I_n$. Columns are the right singular vectors of A .

— $\Sigma \in \mathbb{R}^{n \times m}$:

a rectangular diagonal matrix w/ non-neg entries $\sigma_1, \sigma_2, \dots, 0$ on the diagonal. Singular values of A

— $V \in \mathbb{R}^{m \times m}$

orthogonal matrix, so $V^T V = I_m$. Its columns are the left singular vectors of A

#2 Consider the matrix $A^T A$ a PSD matrix

a) Why $A^T A$ a PSD (positive semi-definite) matrix?

For any vector $x \in \mathbb{R}^m$

$$x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 \geq 0 \Rightarrow$$

$A^T A$ is positive semidefinite (PSD)

b) Use the SVD of A to find the eigendecomposition of $A^T A$

Using $A = U \Sigma V^T$

$$\Rightarrow A^T A = (U \Sigma V^T)^T (U \Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V \Sigma^T \Sigma V^T$$

eigendecomposition of $A^T A$
 — eigenvectors are columns of V
 — eigenvalues are diagonal entries of $\Sigma^T \Sigma$

c) How are the eigenvalues of $A^T A$ related to singular values of A ?

The eigenvalues of $A^T A$ are the squares of the singular values of A :

$$\lambda_i = \sigma_i^2$$

#3 Consider the matrix AA^T and answer the previous questions by replacing $A^T A$ w/ AA^T

$$AA^T = U \Sigma V^T V \Sigma^T U^T = U \Sigma \Sigma^T U^T$$

→ AA^T is PSD because $x^T AA^T x = \|A^T x\|^2 \geq 0$

→ Eigenvectors are columns of U

→ Eigenvalues are again σ_i^2

⇒ AA^T has the same nonzero eigenvalues as $A^T A$

#4 How are the ranks of A , $A^T A$, AA^T related, and how can one use the above to verify?

$$\text{rank}(A) = \text{rank}(A^T A) = \text{rank}(AA^T)$$

- the rank of a matrix equals the # nonzero singular values
- eigenvalues of $A^T A$ and AA^T are σ_i^2
- squaring does not change whether values are zero/nonzero

Code section Explanation

replace image matrix with a lower-rank approx
built from SVD $\rightarrow$ keep only the first k singular values +
matching singular vectors

SVD stores image more efficiently because instead of
keeping every pixel entry, we store only the truncated
factors U_k, Σ_k, V_k^T . If k is smaller than the image
dimensions, the reconstructed image still looks similar $\rightarrow$
uses fewer #s

$$A = U \Sigma V^T$$

U : column behaviors $\rightarrow$ left singular vectors
 $\rightarrow$ the basis patterns image is projected onto

Σ : strength of those behaviors $\rightarrow$ diagonal matrix of singular values
 $\sigma_1, \sigma_2, \dots$ sorted from largest to
smallest. Each σ_i tells you how
important that corresponding
pattern is

V^T : set of row behaviors $\rightarrow$ right singular vectors

